

# Honors Analysis I

## Recitation 4 Notes

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Quiz 4 covering Homework 4 will be next week! Today I can either go through the midterm solutions or talk about metric spaces. These notes will cover the content on metric spaces. I'll write up solutions to the midterm separately.

### 1. CONCEPT REVIEW

A *metric space* is a tuple  $(X, d)$  of a set  $X$  and a *metric*  $d: X \times X \rightarrow [0, \infty)$  which satisfies

- (1)  $d(x, y) = 0$  if and only if  $x = y$ ,
- (2)  $d(x, y) = d(y, x)$  for all  $x, y \in X$ ,
- (3)  $d(x, y) \leq d(x, z) + d(z, y)$  for all  $x, y, z \in X$ .

An *open ball of radius  $r > 0$  at  $x \in X$*  is the set

$$B(x, r) := \{y \in X : d(x, y) < r\}.$$

A *closed ball of radius  $r > 0$  at  $x \in X$*  is the set

$$\overline{B}(x, r) := \{y \in X : d(x, y) \leq r\}.$$

A metric space is called *complete* if all Cauchy sequences converge to some limit in the space. Let's do some examples:

**Example 1.1.** If  $d_1, d_2$  are two given metrics on a set  $X$ , decide whether the following are metrics on  $X$ :

- (a)  $d_1 + d_2$ ,
- (b)  $\max(d_1, d_2)$ .

*Solution:*

(a) Non-negativity is immediate from non-negativity of  $d_1$  and  $d_2$ . If  $x = y$  then  $d_1(x, x) + d_2(x, x) = 0$ . Conversely, if  $d_1(x, y) + d_2(x, y) = 0$ , then  $d_1(x, y) = -d_2(x, y) \leq 0$ , so in fact  $d_1(x, y) = 0$ , hence  $x = y$ . Symmetry is also immediate from symmetry of  $d_1, d_2$ . Given  $x, y, z \in X$ , we compute

$$\begin{aligned} d_1(x, y) + d_2(x, y) &\leq d_1(x, z) + d_1(z, y) + d_2(x, z) + d_2(z, y) \\ &= (d_1(x, z) + d_2(x, z)) + (d_1(z, y) + d_2(z, y)), \end{aligned}$$

so the triangle inequality holds. Thus,  $d_1 + d_2$  is a metric.

(b) As in (a), non-negativity and symmetry are immediate. If  $x = y$ , then

$$\max(d_1(x, y), d_2(x, y)) = \max(0, 0) = 0.$$

If instead  $\max(d_1(x, y), d_2(x, y)) = 0$ , then  $d_1(x, y) = 0$  or  $d_2(x, y) = 0$ . so that  $x = y$ . For the triangle inequality, given  $x, y, z \in X$ , we compute WLOG

$$\begin{aligned}\max(d_1(x, y), d_2(x, y)) &= d_1(x, y) \leq d_1(x, z) + d_1(z, y) \\ &\leq \max(d_1(x, z), d_2(x, z)) + \max(d_1(z, y), d_2(z, y)).\end{aligned}$$

Thus,  $\max(d_1, d_2)$  is a metric. □

**Example 1.2.** Consider the metric space  $(\mathbb{R}^n, d)$  where  $d$  is the usual Euclidean metric. Let  $e_i$  denote the  $i$ -th standard basis vector.

(a) Verify  $d(e_i, e_j) = \sqrt{2}$  when  $i \neq j$ ,

(b) Prove that each  $e_i$  is contained in exactly one ball in the collection

$$\mathcal{C} := \{B(e_i, \sqrt{2})\}_{i=1}^n \cup \{B(-e_i, \sqrt{2})\}_{i=1}^n,$$

(c) Prove that each  $x \in B(0, 1)$  is contained in at least one of the balls in  $\mathcal{C}$ .

*Solution:*

(a) If  $i \neq j$ ,

$$(e_i - e_j)_k = \begin{cases} 1 & k = i \\ -1 & k = j \\ 0 & \text{otherwise} \end{cases}.$$

Thus,

$$d(e_i, e_j) = |e_i - e_j| = \sqrt{\sum_{k=1}^n (e_i - e_j)_k^2} = \sqrt{2}.$$

(b) Since  $d(e_i, e_i) = 0$ , we immediately have  $e_i \in B(e_i, \sqrt{2})$ . It remains to check that  $e_i$  is in no other ball in the collection  $\mathcal{C}$ . If  $i \neq j$ , then  $d(e_i, e_j) = \sqrt{2}$  by (a), and hence  $e_i \notin B(e_j, \sqrt{2})$ . Likewise a similar computation as in (a) shows that  $d(e_i, -e_j) = \sqrt{2}$ , so that  $e_i \notin B(-e_j, \sqrt{2})$ . Finally, observe that

$$d(e_i, -e_i) = \sqrt{(1 - (-1))^2} = 2 > \sqrt{2},$$

so that  $e_i \notin B(-e_i, \sqrt{2})$ .

(c) Suppose there is  $x \in B(0, 1)$  that is contained in no ball of  $\mathcal{C}$ . Then

$$d(x, e_i), d(x, -e_i) \geq \sqrt{2}$$

for each  $i = 1, \dots, n$ . Then, in particular, we have

$$2 \leq d(x, e_1)^2 = |x - e_1|^2 = |x|^2 - 2|x_1| + 1.$$

Therefore,  $1 > |x|^2 \geq 1 + 2|x_1| \geq 1$ , a contradiction. Hence,  $x$  is contained in some ball in  $\mathcal{C}$ . □

**Example 1.3.** Let  $X = \ell^1$  denote the set of real sequences  $\{a_n\}_n$  for which  $\sum_{n=1}^{\infty} |a_n| < \infty$ . Let  $d_1$  and  $d_\infty$  be metrics on  $X$  given by

$$d_1(x, y) = \sum_{n=1}^{\infty} |x_n - y_n|, \quad d_\infty(x, y) = \sup_{n \in \mathbb{N}} |x_n - y_n|$$

for every  $x = \{x_n\}_n, y = \{y_n\}_n \in X$ .

(a) Confirm these are both metrics.

(b) Show there are sequences  $\{x^{(k)}\}_k$  in  $X$  which converge in  $(X, d_\infty)$  but fail to converge in  $(X, d_1)$ .

(c) Show that if  $\{x^{(k)}\}_k$  converges in  $(X, d_1)$ , then it automatically converges in  $(X, d_\infty)$ .

*Solution:*

(a) This is left as an exercise.

(b) Define a sequence  $\{x_k\}_k \subseteq X$  by

$$x_n^{(k)} := \begin{cases} 1/k & n \leq k \\ 0 & n > k, \end{cases}$$

i.e.,

$$x^{(k)} = (\underbrace{1/k, 1/k, \dots, 1/k}_{k \text{ times}}, 0, \dots).$$

Then

$$\sum_{n=1}^{\infty} |x_n^{(k)}| = \underbrace{1/k + \dots + 1/k}_{k \text{ times}} = 1 < \infty,$$

so that  $x^{(k)} \in X$  for every  $k \in \mathbb{N}$ . We claim that  $x^{(k)} \rightarrow 0$  in  $(X, d_\infty)$ . Indeed,

$$\sup_{n \in \mathbb{N}} |x_n^{(k)}| = 1/k \rightarrow 0 \text{ as } k \rightarrow \infty.$$

To show non-convergence in  $(X, d_1)$ , suppose instead  $x^{(k)} \rightarrow x$  as  $k \rightarrow \infty$ . If  $x \neq 0$ , then there is  $m \in \mathbb{N}$  such that  $x_m \neq 0$ . But then

$$|x_m - x_m^{(k)}| \leq \sum_{n=1}^{\infty} |x_n - x_n^{(k)}| \rightarrow 0 \text{ as } k \rightarrow \infty,$$

a contradiction, since  $x_m^{(k)} \rightarrow 0$  as  $k \rightarrow \infty$ . Thus,  $x = 0$ , which contradicts

$$\sum_{n=1}^{\infty} |x_n^{(k)}| = 1$$

for all  $k \in \mathbb{N}$ .

(c) Suppose  $x^{(k)} \rightarrow x$  in  $(X, d_1)$ . Then

$$|x_n^{(k)} - x_n| \leq \sum_{n=1}^{\infty} |x_n^{(k)} - x_n|$$

Taking the supremum over all  $n \in \mathbb{N}$  on the LHS and then taking the limit  $k \rightarrow \infty$  concludes the proof.

□

**Example 1.4.** *Show that the metric space  $(X, d)$  is complete if and only if the metric space  $(X, \tilde{d})$  is complete, where  $\tilde{d} = d/(1 + d)$ .*

*Solution:*

Assume first that  $(X, d)$  is complete. Let  $\{x_n\}_n$  be Cauchy in  $(X, \tilde{d})$ . Given  $\epsilon > 0$ , let  $\delta = \epsilon/(1 + \epsilon)$  and choose  $N \in \mathbb{N}$  such that for all  $m, n \geq N$  we have

$$\frac{d(x_n, x_m)}{1 + d(x_n, x_m)} = \tilde{d}(x_n, x_m) < \delta = \frac{\epsilon}{1 + \epsilon}.$$

Rearranging and canceling terms, we obtain  $d(x_n, x_m) < \epsilon$ . Thus,  $x_n \rightarrow x$  in  $(X, d)$  for some  $x \in X$ . Now the observation that  $\tilde{d} \leq d$  concludes the proof that  $x_n \rightarrow x$  in  $(X, \tilde{d})$ .

We leave the other direction as an exercise.

□